

because it is the best thing for them to do (it is mathematically “optimal”). The opposite is a *positive* science, which is based on how people actually are observed to behave. In spite of economists’ envy of physicists, physics is an inherently positive science while economics, particularly microeconomics and financial economics, is predominantly a normative one.

NEUROBIOLOGY

The soft sciences of psychology and economics have cheated us on occasions in the past. How? Economics has produced laughable ideas, ideas that evaporate once one changes the assumptions a little bit. It seems difficult to take sides with bickering economists trading often-incomprehensible arguments (even to economists). Biology and medicine, on the other hand, rank higher in scientific firmness; like *true* sciences, they can explain things while at the same time being subjected to falsification. They are both positive and their theories are *better* theories, that is, more easily testable. The good news is that neurologists are starting to confirm these results, with what is called environment mapping in the brain, by taking a patient whose brain is damaged in one single spot (say, by a tumor or an injury deemed to be local) and deducing by elimination the function performed by such part of the anatomy. This isolates the parts of the brain that perform the various functions. The Kahneman and Tversky results thus found a *terra firma* with the leaps in our knowledge obtained through behavioral genetics and, further, plain medicine. Some of the physiology of our brain makes us perceive things and behave in a given manner. We are, whether we like it or not, prisoners of our biology.

Researchers in evolutionary psychology provide convincing reasons for these biases. We have not had the incentive to develop an ability to understand probability because we did not have to do so – but the more profound reason is that we are not designed to understand things. We are built only to survive and procreate. To survive, we need to overstate some probabilities, such as those that can affect our survival. For instance, those whose brain imparted higher odds to dangers of death, in other words the paranoid, survived and gave us their genes (provided

such paranoia did not come at too high a cost, otherwise it would have been a drawback). Our brain has been wired with biases that may hamper us in a more complex environment, one that requires a more accurate assessment of probabilities.

The story of these biases is thus being corroborated by the various disciplines; the magnitude of the perceptual distortions makes us less than rational, in the sense of both having coherent beliefs (i.e. free of logical contradictions) and acting in a manner compatible with these beliefs.

EXAMPLES OF BIASES IN UNDERSTANDING PROBABILITY

I found in the behavioral literature at least 40 damning examples of such acute biases. Below is the account of a well-known test, and an embarrassing one for the medical profession. The following quiz was given to medical doctors (which I borrowed from the excellent Deborah Bennett’s *Randomness*).

*A test of a disease presents a rate of 5% false positives. The disease strikes 1/1,000 of the population. People are tested at random, regardless of whether they are suspected of having the disease. A patient’s test is positive. What is the probability of the patient being stricken with the disease?*¹⁰

Most doctors answered 95%, simply taking into account the fact that the test has a 95% accuracy rate. The answer is the conditional probability that the patient is sick and the test shows it – close to 2%. Less than one in five professionals got it right.

I will simplify the answer. Assume no false negatives. Consider that out of 1,000 patients who are administered the test, one will be expected to be afflicted with the disease. Out of a population of the remaining 999 healthy patients, the test will identify about 50 with the disease (it is 95% accurate). The correct answer should be that the probability of being afflicted with the disease for someone selected at random who presented a positive test is the following ratio: